Heart Disease Classification: A Comparative Machine Learning Study

AASD 4001 – Mathematical Concepts for Machine Learning

School of Computer Technology — Term Project Report

Group 4 (5-person team)

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1. Problem Statement

Heart disease remains one of the leading causes of death worldwide. Early and accurate identification of at-risk individuals is critical for timely intervention and improved patient outcomes. This project applies five supervised machine learning classification algorithms to a real-world heart disease dataset to determine how effectively each model can predict whether a patient has heart disease, and to understand the trade-offs involved in tuning and feature selection.

A key challenge in this domain is class imbalance: the dataset contains significantly more patients without heart disease than with it. This means that standard accuracy metrics are misleading — a model that predicts "no heart disease" for every patient would achieve 80% accuracy while catching zero actual cases. Accordingly, our primary evaluation metric throughout this report is Recall (Class 1), which measures what fraction of true heart disease cases the model correctly identifies.

2. Dataset Description

The dataset used is a heart disease classification dataset containing 10,000 patient records and 21 raw features. After preprocessing (described in Section 3), 15 encoded features were used as model inputs. The target variable is Heart Disease Status (binary: Yes/No, encoded as 1/0).

2.1 Feature Summary

Feature	Type	Description
Age	Numeric	Patient age in years
Gender	Categorical	Male / Female
Blood Pressure	Numeric	Systolic blood pressure (mmHg)
Cholesterol Level	Numeric	Total cholesterol level (mg/dL)
Smoking	Categorical	Smoker (Yes/No)
Family Heart Disease	Categorical	Family history of heart disease (Yes/No)
Diabetes	Categorical	Diabetic (Yes/No)
BMI	Numeric	Body Mass Index
High Blood Pressure	Categorical	Clinically diagnosed (Yes/No)
Low HDL Cholesterol	Categorical	Low HDL ("good") cholesterol (Yes/No)
High LDL Cholesterol	Categorical	High LDL ("bad") cholesterol (Yes/No)
Triglyceride Level	Numeric	Blood triglyceride level (mg/dL)
Fasting Blood Sugar	Numeric	Fasting glucose level (mg/dL)
CRP Level	Numeric	C-reactive protein (inflammation marker, mg/L)
Homocysteine Level	Numeric	Amino acid linked to cardiovascular risk (µmol/L)
Heart Disease Status	Categorical (Target)	Presence of heart disease (Yes/No)

2.2 Class Distribution

The dataset is imbalanced: approximately 80% of patients do not have heart disease (Class 0) and 20% do (Class 1). This imbalance directly influenced our choice of evaluation metric (recall) and model configuration (class_weight='balanced').

2.3 Exploratory Data Analysis (EDA)

Three types of visualizations were produced to understand feature distributions and their relationship to the target variable before modeling.

Figure 1 — Correlation Heatmap: A full correlation matrix including the target variable was computed on the preprocessed dataset. Key findings include that most features show weak individual correlations with the target (coefficients in the 0.01–0.08 range), consistent with the difficulty of the classification task. This low correlation profile suggests the dataset's predictive signal is distributed across many features rather than concentrated in a few — which favors ensemble methods like Random Forest over simple linear models. Notably, no single feature strongly drives the outcome, reinforcing the need for multi-feature models.

Figure 2 — Stratified Boxplots: For each of the eight numeric features (Age, Blood Pressure, Cholesterol Level, BMI, Triglyceride Level, Fasting Blood Sugar, CRP Level, Homocysteine Level), boxplots were generated separately for patients with and without heart disease. The median values and interquartile ranges for both classes overlap substantially across most features, confirming the weak individual discriminative power seen in the correlation heatmap. Features like Age, CRP Level, and Homocysteine Level show slightly more separation between classes, suggesting they carry marginally more predictive signal.

Figure 3 — Stacked Bar Charts: Each categorical feature (Gender, Smoking, Family History, Diabetes, High Blood Pressure, Low HDL Cholesterol, High LDL Cholesterol) was visualized as a stacked proportion bar chart showing the percentage breakdown of heart disease cases within each category. Patients with a family history of heart disease showed a notably higher proportion of positive cases. High Blood Pressure and Low HDL Cholesterol also showed elevated heart disease proportions, consistent with established clinical risk factors. Smoking and Diabetes showed a smaller differentiation — supporting the feature ablation finding that these features added limited unique predictive signal.

EDA Takeaway: The data does not contain any single dominant predictor. The class imbalance (80/20 split) combined with low individual feature correlations means all models must work harder to detect the minority class — making class rebalancing strategies (see Section 7) especially important.

3. Data Pre-processing and Cleaning

3.1 Feature Removal

Five features were removed prior to modeling due to high rates of missing values or low relevance to the prediction task:

- Exercise Habits
- Alcohol Consumption (2,586 missing values out of 10,000)
- Stress Level
- Sleep Hours
- Sugar Consumption

3.2 Missing Value Imputation

After feature removal, remaining missing values were handled as follows:

- Numeric columns (Age, Blood Pressure, Cholesterol Level, BMI, Triglyceride Level, Fasting Blood Sugar, CRP Level, Homocysteine Level): filled with the column median.
- Categorical columns (Gender, Smoking, Family Heart Disease, Diabetes, High Blood Pressure, Low HDL Cholesterol, High LDL Cholesterol): filled with the column mode.

After imputation, all columns had zero missing values.

3.3 Encoding Categorical Variables

All categorical features were converted to numeric using pd.get_dummies(drop_first=True), resulting in binary (0/1) encoded columns for each categorical variable. Boolean columns were subsequently cast to integers.

3.4 Train/Test Split and Scaling

The dataset was split into 80% training (8,000 samples) and 20% testing (2,000 samples) using stratify=y to preserve class balance. All features were scaled using StandardScaler, fitted exclusively on training data to prevent data leakage.

Final shape: X_train (8000, 15), X_test (2000, 15).

4. Classification Models — Baseline Results

Five classification models were implemented and evaluated. Each model was configured with class_weight='balanced' where applicable to account for class imbalance.

4.1 Decision Tree

A Decision Tree classifier was trained with max_depth=5 and random_state=42.

Metric	Class 0 (No Disease)	Class 1 (Heart Disease)	Overall
Precision	0.80	0.22	—
Recall	0.99	0.01	—
F1-Score	0.89	0.02	—
Accuracy	—	—	0.80

The Decision Tree achieved 80% accuracy but an alarmingly low recall of 0.01 for heart disease cases — catching only 1% of actual positive cases. This illustrates the danger of relying on accuracy alone with imbalanced datasets.

4.2 Logistic Regression

A Logistic Regression model was trained with max_iter=1000 and class_weight='balanced'.

Metric	Class 0 (No Disease)	Class 1 (Heart Disease)	Overall
Precision	0.80	0.20	—
Recall	0.54	0.46	—
F1-Score	0.64	0.28	—
Accuracy	—	—	0.52

Logistic Regression, with balanced class weights, achieved a recall of 0.46 — catching 46% of actual heart disease patients, at the cost of lower overall accuracy (52%).

4.3 Random Forest

A Random Forest classifier was initialized with class_weight='balanced' and a custom probability threshold of 0.2 to increase sensitivity.

Metric	Class 0 (No Disease)	Class 1 (Heart Disease)	Overall
Precision	0.79	0.19	—
Recall	0.51	0.46	—
F1-Score	0.62	0.27	—
Accuracy	—	—	0.50

The baseline Random Forest achieved a recall of 0.46, consistent with Logistic Regression. The ensemble approach showed robustness to noise in individual features.

4.4 SGD — Stochastic Gradient Descent

An SGD classifier was trained using stochastic gradient descent optimization and class_weight='balanced' to balance class weight.

Metric	Class 0 (No Disease)	Class 1 (Heart Disease)	Overall
Precision	0.80	0.20	—
Recall	0.28	0.72	—
F1-Score	0.41	0.31	—
Accuracy	—	—	0.37

The SGD model achieved a recall of 0.72 for heart disease cases, the highest sensitivity among the reported results, but with a low overall accuracy of 0.37. This indicates that SGD was more aggressive in identifying positive cases, catching many true heart disease patients while also producing more false positives.

4.5 SVM — Support Vector Machine

An SVM classifier with an RBF kernel, class_weight='balanced', was trained on the dataset.

Metric	Class 0 (No Disease)	Class 1 (Heart Disease)	Overall
Precision	0.80	0.20	—
Recall	0.59	0.42	—
F1-Score	0.68	0.27	—
Accuracy	—	—	0.56

SVM achieved a recall of 0.42 and the highest baseline accuracy (56%) among the balanced models, reflecting the kernel method's effectiveness at finding non-linear boundaries.

5. Hyperparameter Tuning via GridSearchCV

GridSearchCV was applied to all five models with scoring='recall' to optimize for the detection of heart disease cases. Cross-validation used 3–5 folds depending on the model.

5.1 Summary of Tuned Results

Model	Best Parameters	Recall (Baseline)	Recall (Tuned)
Decision Tree	max_depth=5, tuned via GridSearch	0.01	0.43
Logistic Regression	C=0.01, penalty='l2', solver='lbfgs'	0.46	0.46
Random Forest	max_depth=5, n_estimators=50	0.46	0.33
SGD	Tuned parameters	0.72	0.52
SVM (SVC)	C=10, gamma='scale', kernel='linear'	0.42	0.47

5.2 Observations

GridSearchCV improved accuracy across most models, but the relationship between tuning and recall was not always positive. Notably:

- Random Forest: Recall decreased from 0.46 to 0.33 after tuning, despite optimizing for recall. The tuned model (max_depth=5, 50 estimators) is more structurally constrained and learned to classify more conservatively, reducing false positives but also reducing true positive detection.
- Decision Tree: Recall improved significantly, from 0.01 to 0.43, demonstrating that tree-depth constraints dramatically affect sensitivity on imbalanced data.
- SVM: The linear kernel (C=10) outperformed the RBF baseline in recall, suggesting the data may not have strong non-linear structure separating the two classes.
- Logistic Regression: Tuning had minimal effect — the baseline and tuned results were nearly identical, suggesting the model was already near its representational limit.

6. Feature Ablation Study

Following the hypothesis that Smoking and Diabetes — while clinically significant risk factors — may be informationally redundant with other features already in the dataset, these two features were removed and all five models were re-evaluated.

6.1 Ablation Results — All Models

Model	Recall (Tuned)	Recall (After Ablation)	Change
Decision Tree	0.43	0.37	-0.06 from tuned
Logistic Regression	0.46	0.43	-0.03 from tuned
Random Forest	0.33	0.33	No change
SGD	0.52	0.72	+0.20 from tuned
SVM (SVC)	0.47	0.47	No change

6.2 Observations

- Random Forest and SVM: Showed virtually no change in performance after feature removal, confirming that Smoking and Diabetes were not contributing unique signal — the remaining features captured the same information.
- SGD: Exhibited a notable increase in recall (0.52 → 0.72) after feature removal. This is a counterintuitive result, suggesting that for SGD, these features may have introduced noise or created a conflicting gradient signal during training.
- Decision Tree and Logistic Regression: Showed minor decreases in recall (-0.03 to -0.06), suggesting these features had marginal but measurable contributions for those algorithms.
- Overall: The ablation study demonstrated strong model robustness. No model collapsed after removing two features, validating that the remaining feature set carries sufficient predictive signal.

Key Finding: The consistency of performance after feature removal suggests that Smoking and Diabetes are correlated with other features already present (e.g., BMI, Blood Pressure, Triglycerides). In a real clinical deployment, domain expertise should guide any feature removal decisions.

7. Class Rebalancing with SMOTE

Following professor feedback, Synthetic Minority Oversampling Technique (SMOTE) was applied to the training data to address the 80/20 class imbalance. SMOTE generates synthetic samples for the minority class (Heart Disease = Yes) by interpolating between existing minority class observations, rather than simply duplicating them.

Methodology: SMOTE was applied only to the training set after scaling to prevent data leakage into the test set. The training set grew from 8,000 to approximately 12,800 samples (6,400 per class after resampling). The test set remained unmodified at 2,000 samples with the original 80/20 distribution.

7.1 SMOTE Results — All 5 Models

Model	Recall (Before SMOTE)	Recall (After SMOTE)	Change
Decision Tree	0.01	0.66	+0.65
Logistic Regression	0.46	0.42	-0.04
Random Forest	0.46	0.47	+0.01
SGD	0.52	0.45	-0.07
SVM	0.42	0.46	+0.04

Note: figures reflect the SMOTE comparison run in the accompanying Jupyter notebook; see the notebook's before/after recall chart for the full side-by-side visualization.

7.2 Why SMOTE Helps

With the original 80/20 imbalance, models see four times as many examples of “no heart disease” as “heart disease” during training, creating a systematic bias toward the majority class. SMOTE counteracts this by expanding the minority class training examples, giving models more decision boundary exposure for positive cases. Unlike simple duplication (oversampling), SMOTE creates novel synthetic points, which reduces the risk of overfitting to the minority class.

7.3 Trade-offs

While SMOTE is generally expected to improve recall, our results showed that the impact varies by model architecture. It dramatically improved the sensitivity of the Decision Tree (recall increased from 0.01 to 0.66), but it actually harmed the recall of linear models like Logistic Regression and SGD. Furthermore, SMOTE typically reduces precision — the model becomes more willing to flag patients as positive, catching more true positives but also generating more false positives. In a medical screening context, this is often the preferred trade-off: it is more acceptable to flag a healthy patient for follow-up than to miss a patient with heart disease. The optimal balance depends on clinical deployment context.

8. Summary of Findings

8.1 Overall Model Performance Comparison

Model	Recall (Baseline)	Recall (Tuned)	Recall (Ablation)
Decision Tree	0.01	0.43	0.37
Logistic Regression	0.46	0.46	0.43
Random Forest	0.46	0.33	0.33
SGD	0.72	0.52	0.72
SVM	0.42	0.47	0.47

8.2 Key Takeaways

- Accuracy is a misleading metric on this dataset. The Decision Tree's 80% baseline accuracy masked a near-zero recall, proving that an unbalanced metric rewards models that predict the majority class.
- SGD achieved the highest recall after feature ablation (0.72), making it the most sensitive model for detecting true heart disease cases in this study.
- SVM was the most consistently balanced model across all three conditions, with recall moving 0.42 → 0.47 → 0.47.
- GridSearchCV does not guarantee recall improvement. In the Random Forest case, optimizing hyperparameters for recall during cross-validation actually reduced recall on the final test set — illustrating the precision-recall tradeoff and the difference between CV performance and hold-out performance.
- Feature redundancy is real. Removing Smoking and Diabetes had minimal impact across most models, confirming that correlated features do not always add unique predictive value.
- SMOTE addresses class imbalance at the data level rather than the algorithm level, providing a complementary approach to class_weight='balanced' for improving recall on the minority class.

9. Conclusions

This project applied five machine learning classifiers — Decision Tree, Logistic Regression, Random Forest, SGD, and SVM — to a heart disease dataset with significant class imbalance. Using recall as our primary metric, the following conclusions are drawn:

- No single model achieved dramatically high recall prior to class rebalancing, with the best unassisted result being SGD at 0.72 (after feature ablation). EDA confirmed that the dataset's predictive signal is diffuse — no single feature is strongly predictive — which inherently limits classifier performance.
- Hyperparameter tuning via GridSearchCV is valuable but does not guarantee monotonic improvement in the target metric — especially when CV fold performance does not transfer linearly to the hold-out test set.
- Class imbalance must be addressed explicitly, either at the algorithm level (class_weight='balanced') or at the data level (SMOTE). However, we found that SMOTE does not universally improve recall; while it proved highly effective for the Decision Tree model, it actually reduced the sensitivity of linear classifiers (Logistic Regression and SGD).
- Feature ablation revealed model robustness and informational redundancy — removing Smoking and Diabetes did not harm most models and in some cases (SGD) improved recall, consistent with the EDA finding that these features have low unique discriminative power.
- EDA with stratified visualizations (boxplots, stacked bar charts, correlation heatmap) provided critical context for understanding why all models face a fundamentally challenging classification problem — one where the signal is weak and the classes are heavily imbalanced.

9.1 Future Directions

- SMOTE with GridSearchCV: Combining SMOTE resampling with hyperparameter tuning in a pipeline would allow systematic optimization of recall on properly balanced training folds.
- Threshold adjustment: Lowering the classification probability threshold (e.g., to 0.3 or 0.2) would bias the model toward sensitivity at the cost of precision — a reasonable trade-off in medical screening.
- Additional features: Clinical features such as ECG results, exercise stress test data, or genetic markers could dramatically improve predictive power given the weak signal in the current feature set.
- Boosting models: Gradient Boosting or XGBoost with built-in class imbalance handling may yield better recall without sacrificing as much precision.

10. References

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